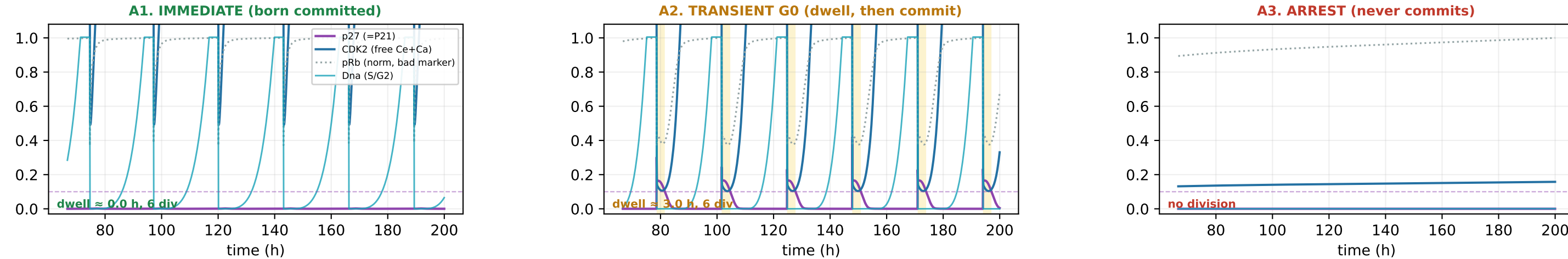
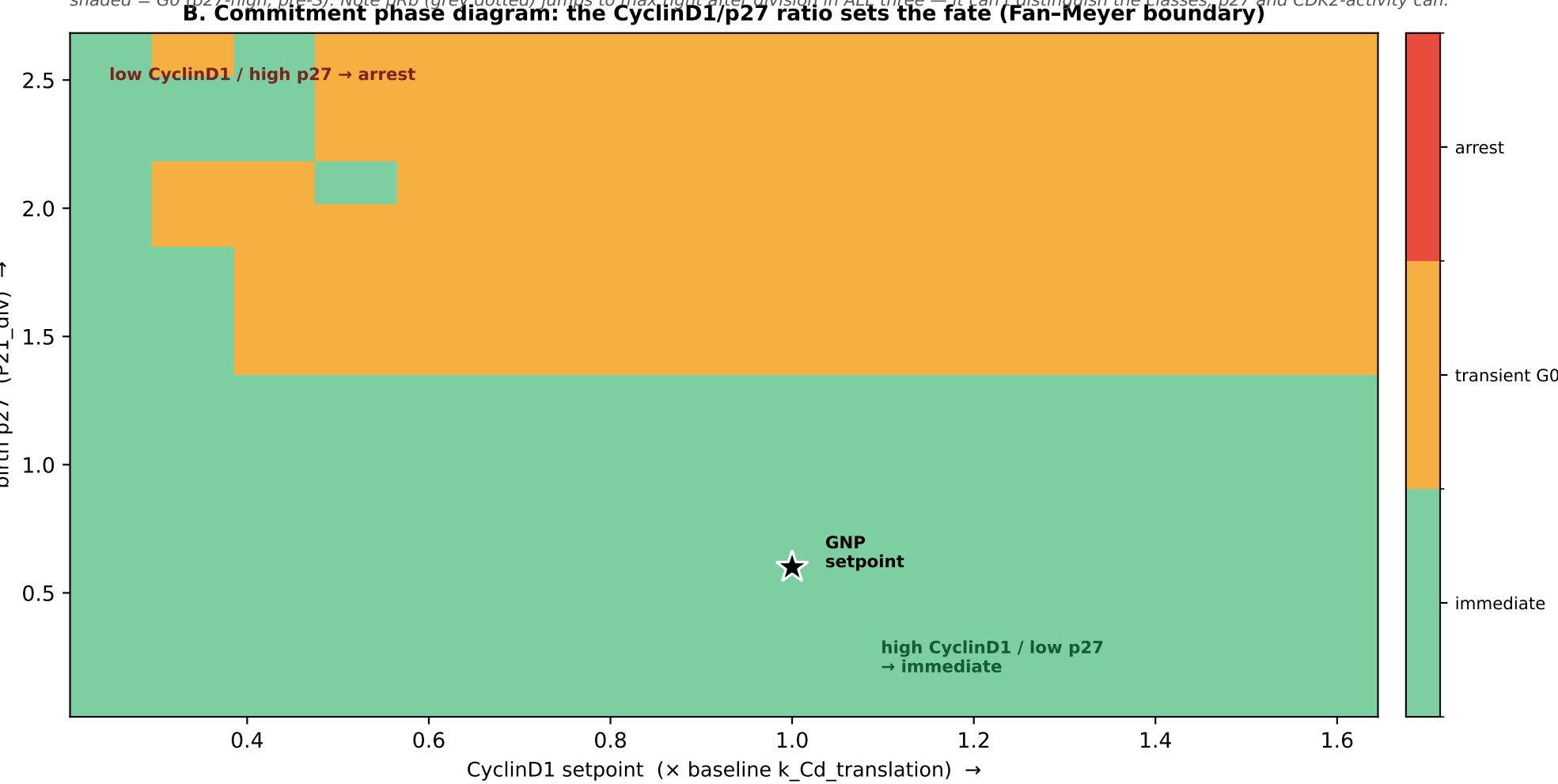


Transient G0 vs immediate G1/early-S entry — definition, mechanism, and CyclinD1/p27 control (v44)

G0 = p27-high / pre-S dwell after birth (NOT pRb, which saturates early). immediate: dwell ≤ 2 h · transient G0: commits after a dwell · arrest: never commits.



shaded = G0 (p27-high, pre-S). Note pRb (grey dotted) jumps to max right after division in ALL three — it can't distinguish the classes; p27 and CDK2-activity can.



DEFINITION (per daughter, at birth):

- born-state = CyclinD1/p27 ratio
(birth p27 = P21_div; CyclinD1 setpoint)
- G0 dwell = time from birth until p27 drops below threshold (Skp2–p27 toggle flips; CDK2 freed). Marked by p27, not pRb.
- IMMEDIATE dwell ≤ 2 h
(Spencer carryover keeps pRb/CycE/low-p27)
 - TRANSIENT G0 dwells, then commits
(grows to M_commit / CyclinD1 rises)
 - ARREST never commits
(CyclinD1 below the threshold)